Modeling with partial differential equations

Exercises – Session 1: Ordinary Differential Equations

Exercise 1 (Duhamel/Variation of Constants for a Linear System)**.** *Consider the ordinary differential equation on* $X\colon [0,T] \to \mathbb{R}^d$*:*

$$\dot{X}(t) = A\,X(t) + B(t), \qquad X(0) = X_0, \tag{1}$$

where $A \in \mathbb{R}^{d\times d}$ *is constant and* $B : [0,T] \to \mathbb{R}^d$ *is continuous.*

(a) *Show that the solution of the homogeneous system* $\dot{X} = AX$ *with* $X(0) = X_0$ *is*

$$X_{\mathrm{hom}}(t) = \mathrm{e}^{tA}\,X_0,$$

where $\mathrm{e}^{tA} := \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k$*. (You may verify that* $\frac{d}{dt}\mathrm{e}^{tA} = A\mathrm{e}^{tA}$*.)*

(b) *Consider* X *the solution of* (1)*, define* $Y(t) := \mathrm{e}^{-tA}X(t)$*. Differentiate* Y *and show that*

$$\frac{d}{dt}\big(\mathrm{e}^{-tA}X(t)\big) = \mathrm{e}^{-tA}B(t).$$

Deduce that, for all $t \in [0,T]$*,*

$$X(t) \;=\; \mathrm{e}^{tA}X_0 \;+\; \int_0^t \mathrm{e}^{(t-s)A}\,B(s)\,ds.$$

(c) ***A simple example.*** *Compute* $X(t)$ *explicitly (without the integral term) in the case* $B(t) = \mathrm{e}^{-\alpha t}\mathbb{1}$*, where* $\mathbb{1} = (1,\dots,1)^\top \in \mathbb{R}^d$*, and* $\alpha \in \mathbb{R}$*, assuming that* $\alpha \notin \sigma(-A)$ *the spectrum of* $-A$*, so that* $A + \alpha I$ *is invertible.*

Exercise 2 (About Lemma 6 (Grönwall's inequality) and applications)**.**

(a) ***(Grönwall)*** *Let* $T > 0$*. Suppose* $u \in C^0([0,T])$ *and, for all* $t \in [0,T]$*,*

$$u(t) \;\le\; \int_0^t a(s)\,u(s)\,ds \;+\; b(t),$$

with $a, b \in C^0([0,T])$ *and* $a \ge 0$*. Show that, for all* $t \in [0,T]$*,*

$$u(t) \;\le\; b(t) \;+\; \int_0^t b(s)\,a(s)\,\exp\Big(\int_s^t a(r)\,dr\Big)\,ds.$$

Hint: *Set* $v(t) := \int_0^t a(s)\,u(s)\,ds$*, and differentiate it.*

(b) ***(Continuous dependence / perturbation estimate)*** *Let* $f : \mathbb{R}^d \to \mathbb{R}^d$ *be continuous and globally* L*-Lipschitz. Let* $x : [0,T] \to \mathbb{R}^d$ *be a* C^1 *function such that*

$$\begin{cases} \dot{x} = f(x), & \text{on } [0,T], \\ x(0) = x_0, \end{cases}$$

and let $y : [0,T] \to \mathbb{R}^d$ be a C^1 function satisfying the perturbed problem

$$\begin{cases} \dot{y} = f(y) + \varepsilon, & \text{on } [0,T], \\ y(0) = y_0, \end{cases}$$

where $\varepsilon : [0,T] \to \mathbb{R}^d$ is continuous. Using question (a), show that for all $t \in [0,T]$,

$$\|x(t) - y(t)\| \;\le\; \mathrm{e}^{LT} \left(\|x_0 - y_0\| + \int_0^T \|\varepsilon(s)\|\, ds \right).$$

Hint: *write an integral equation for $w := x - y$ and apply question (a) with $u(t) = \|w(t)\|$.*

Exercise 3 (Non-uniqueness when f is not Lipschitz: $\dot{x} = \sqrt{|x|}$)**.** *Consider the ordinary differential equation:*

$$\forall t \ge 0,\ \dot{x}(t) = \sqrt{|x(t)|}, \qquad x(0) = 0.$$

(a) *Exhibit two distinct solutions.*

(b) *Explain why this does not contradict the Cauchy-Lipschitz (Picard–Lindelöf) theorem by proving that $f(x) = \sqrt{|x|}$ is* not *locally Lipschitz at $x = 0$.*

(c) *Show that for any $t_0 \ge 0$,*

$$x_c(t) := \begin{cases} 0, & 0 \le t \le t_0, \\ \dfrac{(t - t_0)^2}{4}, & t \ge t_0, \end{cases}$$

is a solution; conclude that there are infinitely many solutions.

Exercise 4. *The logistic equation*

$$\dot{x}(t) \;=\; r\, x(t) \Big(1 - \frac{x(t)}{K} \Big) \tag{2}$$

models the evolution of the density $x(t)$ of a species depending on a limited resource, e.g., sheep on an island, with no predators, but needing to find grazing fields. When the density of the population is low, the per-capita growth rate is close to $r > 0$. When the density increases, the growth rate decreases. It becomes negative if the density is higher than the carrying capacity K of the environment. This equation can be solved explicitly, e.g., by separation of variables, but the aim of this exercise is to show that its qualitative behavior may be understood without having an explicit formula for x.

Let $x_0 \in \mathbb{R}_+$. Let $(J, x(\cdot))$ denote the maximal solution of (2) such that $x(0) = x_0$.

1. *What happens if $x_0 = 0$ or $x_0 = K$?*
2. *Assume $0 < x_0 < K$.*
   (a) *Show that $0 < x(t) < K$ for all $t \in J$ and that $J = \mathbb{R}$.*
   Show that $x(\cdot)$ is strictly increasing.
   (b) *Show that $x(t)$ has a limit in $[x_0, K]$ as $t \to +\infty$ and in $[0, x_0]$ as $t \to -\infty$.*

(c) *Show that if* $x(t) \to x^* \in \mathbb{R}$ *as* $t \to +\infty$, *then* $f(x^*) = 0$, *where* $f(x) = r\,x\big(1 - \frac{x}{K}\big)$.

(d) *Show that* $x(t) \xrightarrow[t\to+\infty]{} K$ *and* $x(t) \xrightarrow[t\to-\infty]{} 0$.

3. *Assume* $x_0 > K$. *Show that* $x(t) > K$ *for all* $t \in J$, *that* $x(\cdot)$ *is decreasing, that* $\sup J = +\infty$ *and that* $x(t) \to K$ *as* $t \to +\infty$.
Show that $x(t) \to +\infty$ *as* $t \to \inf J$.

4. *Assume* $x_0 < 0$ *(a case with limited ecological interest). Show that* $x(t) < 0$ *for all* $t \in J$, *that* $x(\cdot)$ *is decreasing, that* $\inf J = -\infty$ *and* $x(t) \to 0$ *as* $t \to -\infty$.
Show that $x(t) \to -\infty$ *as* $t \to \sup J$.

5. *Sketch the graph of representative solutions and the phase line.*

Exercise 5 (Consistency of some ODE solvers)**.** *Consider the ordinary differential equation*

$$\dot{x}(t) = f(x(t)),$$

where $f : \mathbb{R} \to \mathbb{R}$ *is sufficiently smooth. Let* $\Delta t > 0$ *and* $t_n = n\Delta t$. *We consider the two explicit one-step schemes*

- Explicit Euler*:*
$$x_{n+1} = x_n + \Delta t\, f(x_n)$$
- Heun's method*:*
$$x_{n+1} = x_n + \frac{\Delta t}{2}\big(f(x_n) + f(x_n + \Delta t\, f(x_n))\big)$$

(a) *Verify mathematically that explicit Euler is of order* 1 *and that Heun's method is of order* 2.

Hint*: Use Taylor expansions to evaluate the difference at time* t_{n+1} *between the exact solution* $x(t_{n+1})$ *and the numerical solution* x_{n+1}, *assuming* $x(t_n) = x_n$.

(b) *Verify numerically the orders found in (a). Pick a test problem with a known exact solution, e.g.* $f(x) = \lambda x$ *(so* $x(t) = x_0 \mathrm{e}^{\lambda t}$*) or the logistic right-hand side. On a fixed interval* $[0, T]$, *and for a sequence of decreasing time steps* Δt, *compute the error*

$$E(\Delta t) := \max_{0\le n\le N} \big|x(t_n) - x_n\big| \qquad (\textit{or simply } |x(T) - x_N|),$$

and plot $E(\Delta t)$ *against* Δt *in log–log scale. Check that the slopes are close to* 1 *for explicit Euler and to* 2 *for Heun's method.*

Exercise 6 (A stiff model: explicit vs. implicit Euler)**.** *Consider the scalar ODE*

$$\dot{x}(t) = f(t, x) := \alpha\big(x(t) - \sin t\big) + \cos t, \qquad x(0) = x_0, \tag{3}$$

with a constant parameter $\alpha \in \mathbb{R}$.

(a) The analytical solution. *Find an explicit solution to* (3) *(Duhamel formula or any other technique can be used). Identify the long-time behavior of solutions depending on the sign of* α.

In the following, we assume $\alpha < 0$.

(b) Time discretization (explicit/implicit Euler). *Let* $t_n = n\Delta t$. *Derive (we use the notation* $x_n \approx x(t_n)$*):*

- *the* explicit Euler *scheme;*
- *the* implicit Euler *scheme; solved in closed form for* x_{n+1}.

Why is this model stiff*? One says that a problem is stiff if the stability constraint is more stringent to determine the time step than the precision constraint. This typically occurs in systems with two time scales: on the fast timescale, the system relaxes towards the dynamics with slow time scale. In our context, the fast time scale is related to the parameter* α. *On our example, for* $\alpha < 0$, *solutions relax to* $\sin t$ *with fast time scale* $|\alpha|^{-1}$. *When* $|\alpha| \gg 1$ *there are two widely separated time scales:*

$$\textit{fast} = |\alpha|^{-1}, \qquad \textit{slow/forcing} = 2\pi,$$

so the problem is stiff.

(c) Simulations. *Implement both schemes to compute an approximation over the time interval* $[0, 10]$ *and run them for* $\alpha = -0.5$. *For the explicit scheme, vary* Δt *to see its influence.*

(d) Absolute stability insight. *Use different values of* $\alpha < 0$ *and* Δt *to illustrate the stiffness of the problem. For instance* $(\alpha, \Delta t) = (-5,\, 0.01)$, $(-10,\, 0.18)$, $(-10,\, 0.201)$, $(-10,\, 0.22)$. *Show numerically that explicit Euler is stable only if* $|1 + \alpha\Delta t| < 1$, *whereas implicit Euler is always stable for* $\alpha < 0$.

(e) Order of accuracy. *Choose a time step small enough that the schemes are stable in your setting. Verify numerically that both explicit and implicit Euler are first-order accurate in* Δt.